AI Forward Deployed Engineer

The AI Architect & FDE Course



Week 3 Day 3

3 weeks to Forward Deployed Engineer

Theory Practical Projects

Week 3: Become an AI FDE

AI Solutions Architect	Security, Guardrails, API Gateways	Enterprise Context Layer & Integrations	Production Deployment	Capstone

Week 2: Become an AI Architect — AI ARCHITECT

Architecture, Evals, Observability	Agentic RAG, MCP, Memory	Durable State & Human in the Loop	Agent Teams, Swarms, Fleet	Self-Improvement, RSI

Week 1: Become an AI Engineer — AI ENGINEER

Path to FDE	Frontier LLMs & Agents	Coding Agents & ADLC	Open-Source, Local, Token Cost	Harness Engineering

Taking stock

You can already:

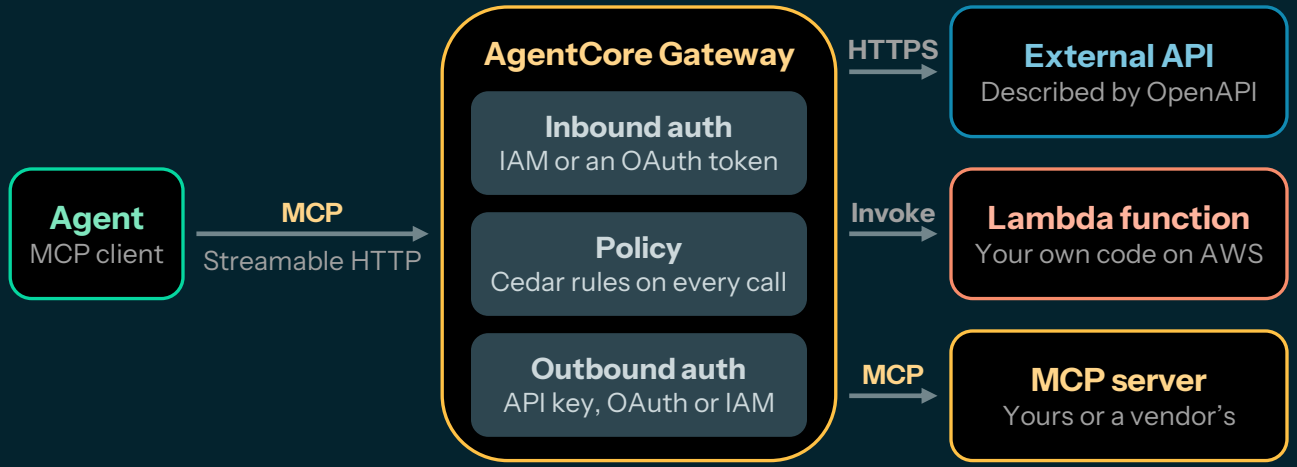
- [x] Deliver solutions that measurably solve business problems with evals and observability
- [x] Create durable, scalable agents with memory and a human in the loop
- [x] Drive an AI implementation from discovery to delivery and maintenance
- [x] Integrate agents with APIs, LLMs and MCP tools with a common Gateway

By the end of today, you'll be able to:

- [ ] Explain the purpose of an enterprise context layer for AI agents
- [ ] Build a context layer from a business's documents and reference data
- [ ] Connect an agent to enterprise systems through an MCP gateway
- [ ] Design an agent's long-term memory with namespaces and strategies

Recap: the Gateway

The agent talks to the Gateway over MCP



The agent sees one list of tools

Where the term came from

From context engineering in 2025 to the enterprise context layer

Context engineering

"the right information and tools, in the right format, at the right time"

Philipp Schmid, June 2025

- Instructions
- User prompt
- Available tools
- Short-term memory
- Long-term memory
- Retrieved information
- Structured output

The enterprise context layer

"the system that turns knowledge, expertise, and norms into machine-usable context for AI"

Atlan, June 2026

- **What the data means**
- **How the work is done**
- **What the agent may see and do**
- **Where it came from**

Most of the pieces are long-standing data governance: data dictionaries, procedure manuals, access controls and lineage

The enterprise context layer

Provide agents with a holistic understanding

"The deductible applies once per claim to Baggage and Personal Effects and to Emergency Medical. It does not apply to the other benefits." Policy wording, section 1

An agent with access to every record can still deduct it from a Trip Delay payout

A context layer gives agents four kinds of context at run time, usually over MCP



What the data means

Data dictionary, glossary and reference data



How the work is done

Procedures and business rules



What the agent may see and do

Permissions and approvals



Where it came from

Source, owner, lineage and review date

Gartner's view June 2026

Semantics
What the data means

Provenance
Where it came from

Operational state
A golden record per customer and product

How Master Data Management Boosts the Context Layer for AI Agents

Where each part lives in UnderCover

We'll build three of these today

What the data means	The data dictionary, the glossary, the plans and reference data	Context layer
How the work is done	The claims manual and the claim rules	Context layer
What the agent may see and do	The Gateway's Policy, the agent's IAM role and the approval limit	Gateway, Day 2
Where it came from	Each file's source system, owner and review date	Context layer

The gateway and three sources of data

Each one has a different owner

	Who writes it	The agent
 **Gateway** Rules for tool calls	**The security team** Policies like the payout limit	**Cannot change it** Every call goes through it
 **Context layer** Data from other systems	**The business teams** Data governance model	**Reads it** As tools through the gateway
 **Memory** The agent's own notes	**The agent** Separate for each customer	**Reads and writes it** With its memory tools
 **Database** The system of record	**The app and the agent** Records and checkpoints	**Reads and writes it** Through the app's API

The context layer market

Most of the big products cover all four parts

	What the data means	How the work is done	What the agent may see and do	Where it came from	Over MCP	Works across
Atlan	●	○	●	●	●	Many platforms
Collibra	●		○	●	●	Many platforms
Alation	●	●	●	●	●	Many platforms
DataHub	●	○	○	●	●	Many platforms
Snowflake Horizon Catalog	●	●	●	●	●	Snowflake
Databricks Unity Catalog	●	●	●	●	○	Databricks
Google Cloud Knowledge Catalog	●	●	●	●	●	Google Cloud
Microsoft Fabric IQ	●	●	●	●	●	Fabric
Palantir Ontology	●	●	●	●	●	Foundry

● Generally available ○ Preview, beta or a narrow version

Checked against vendor docs, October 2026

Memory

Short-term memory is the conversation history; the rest is long-term

The four long-term memory extraction strategies



Semantic
Facts extracted from conversations



Preference
User related information



Summary
Compressed information about what's happened



Episodic
Past task episodes: actions and outcomes

Partitioned by namespace: per customer, per session, per concern

```
/actor/{actorId}/preferences/   and   /actor/{actorId}/session/{sessionId}/summary/
RetrieveMemoryRecords(namespace="/actor/42/preferences/", query="reporting style")
```

A support agent: Semantic holds the customer's setup, Preference their summary-first style, Episodic specific issues & fixes.

The way to find the right memory design: experiment and pick the strategy that performs best against your business metric.

Simpler is usually better: avoid over-engineering unless it solves a specific problem.

Memory products

What each one offers in October 2026

	Managed service	Open source	Facts and preferences	Summaries	Episodes	Knowledge graph	LangGraph integration
AgentCore Memory AWS	●		●	●	●		●
Zep	●	○	●	●		●	●
Mem0	●	●	●	○		○	○
Letta	●	●	●	○			
Cognee	●	●	●		○	●	○
LangGraph store with LangMem	○	●	●		●		●
Memory Bank Google Cloud	●		●				

● Yes ○ Partial: paid plan only, preview or limited

Enterprise data concerns

Ask these questions about your architecture



Systems of record

Where does each record live?

UnderCover
The agent reads policies, customers and claims through the app's API



Read and write access

What can the agent change?

UnderCover
Its only writes are a decision and a payment, both checked by the app



Data boundaries

Where may the data go?

UnderCover
Manual section 11 keeps a customer's data on their own claims and inside the platform



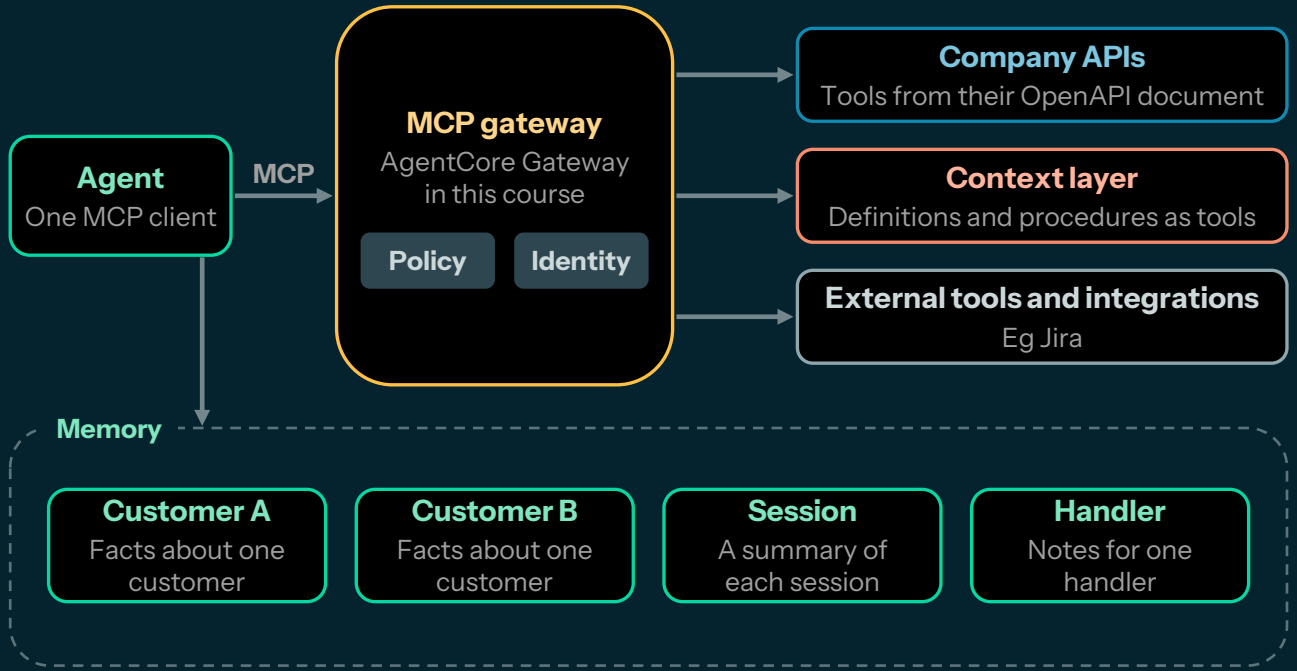
Permissions

Whose access does it use?

UnderCover
The agent has its own IAM role and API key. The Gateway's Policy checks every call

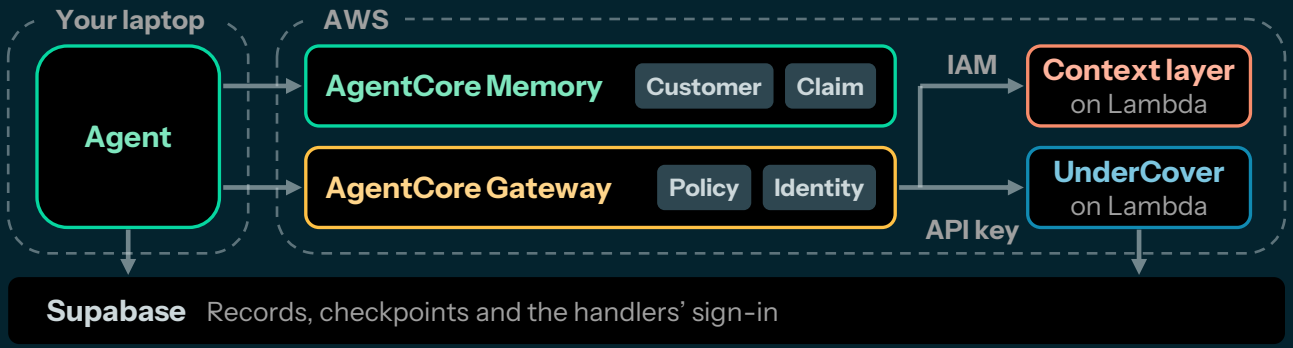
Enterprise MCP architectures

One gateway in front of the company's APIs and its context layer



UnderCover by the end of today

The agent works through the Gateway with context and memory



Agent
Calls every tool through the Gateway

Context layer
Enterprise data behind the Gateway

AgentCore Memory
The agent's memory of each customer

Claim page
Shows each tool call, lookup and memory

The agent moves onto AgentCore Runtime on Day 4

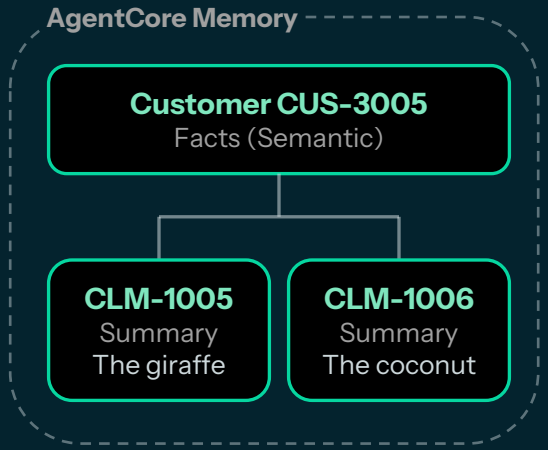
UnderCover's context layer

Each file stands in for another system at the firm

File	What it holds	Where it would come from
Data dictionary	The meaning of each field the agent sees	The data catalog
Glossary	The policy wording's defined terms	The wording, section 12
Plans	Each plan's limits and deductible	Policy administration
Claim rules	Filing window, look-back and repeat claims	Claims operations
Reference data	Outside events, places and products	Outside data providers
Policy wording	The contract with the customer, by section	The legal team
Claims manual	How handlers work a claim, by section	Claims operations

How UnderCover uses memory

The agent remembers each customer across their claims



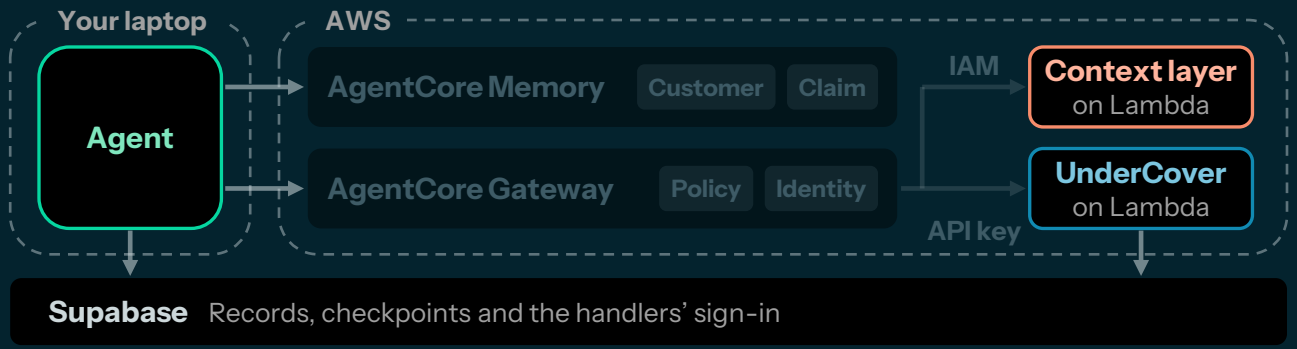
1. **The agent saves each claim's steps**
   As events under the customer and the claim
2. **AgentCore extracts facts and a summary**
   In the background, one strategy each
3. **On the next claim, a tool searches them**
   The agent decides when to look

Oliver Grant's coconut claim in August recalls his giraffe claim from July

Memory calls go straight to AgentCore, limited by the agent's IAM role
Stretch: Episodic memory and a namespace per handler

Upgrading the product

Five changes to the app and the agent



1. **Tools from the Gateway**
   An MCP client signed with SigV4
2. **A context layer behind the Gateway**
   Its own MCP server on Lambda
3. **A complete decision record**
   Checked against manual section 14
4. **Memory of each customer**
   Saved and recalled with memory tools
5. **Every step on the claim page**
   Tools, lookups, memories and refusals

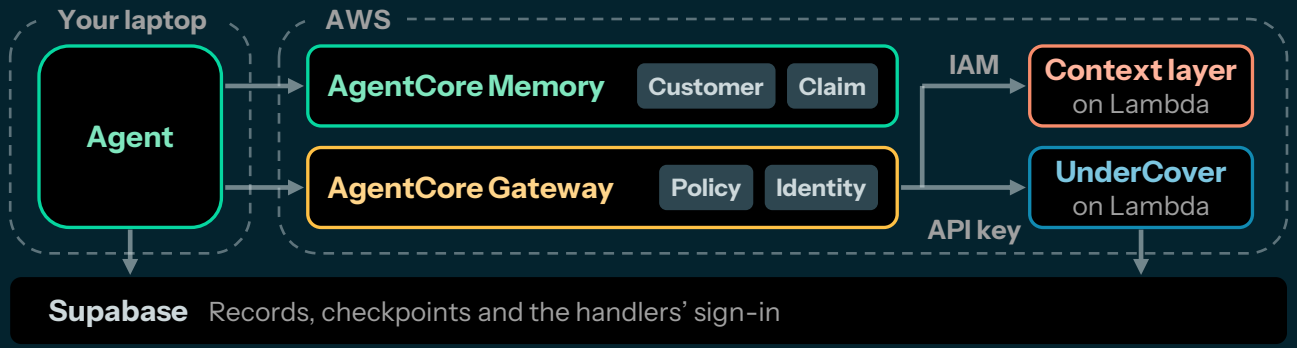
Building it with Claude Code

The change is written up in labs/week3day3_product_CONTEXT.md

1. **Copy the brief into docs**
   `cp labs/week3day3_product_CONTEXT.md docs/CONTEXT.md`
2. **Ask Claude Code for a plan**
   Please plan the context layer changes in docs/CONTEXT.md
3. **Let it build and commit each phase**
   Please implement the plan for the context layer
4. **Or check out my version**
   `git checkout week3day3-start` or `git checkout week3day3-end`

Today's lab

Running UnderCover with its context layer on AWS

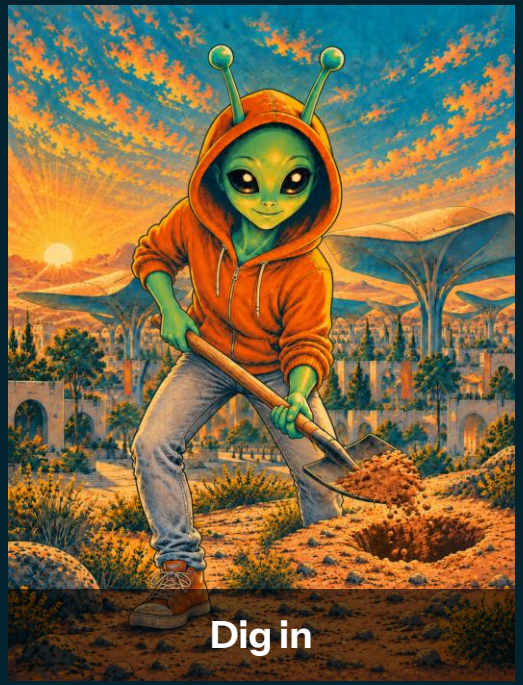


1. **Deploy the new app to Lambda**
   The same function with the new routes
2. **Deploy the context layer**
   With the documents split by section
3. **Recreate the Gateway**
   The app and the context layer as targets
4. **Create the Memory**
   With Summary and Semantic strategies
5. **Run a returning customer's claims**
   The second claim recalls the first
6. **Tear down**
   Delete the Gateway and the Memory

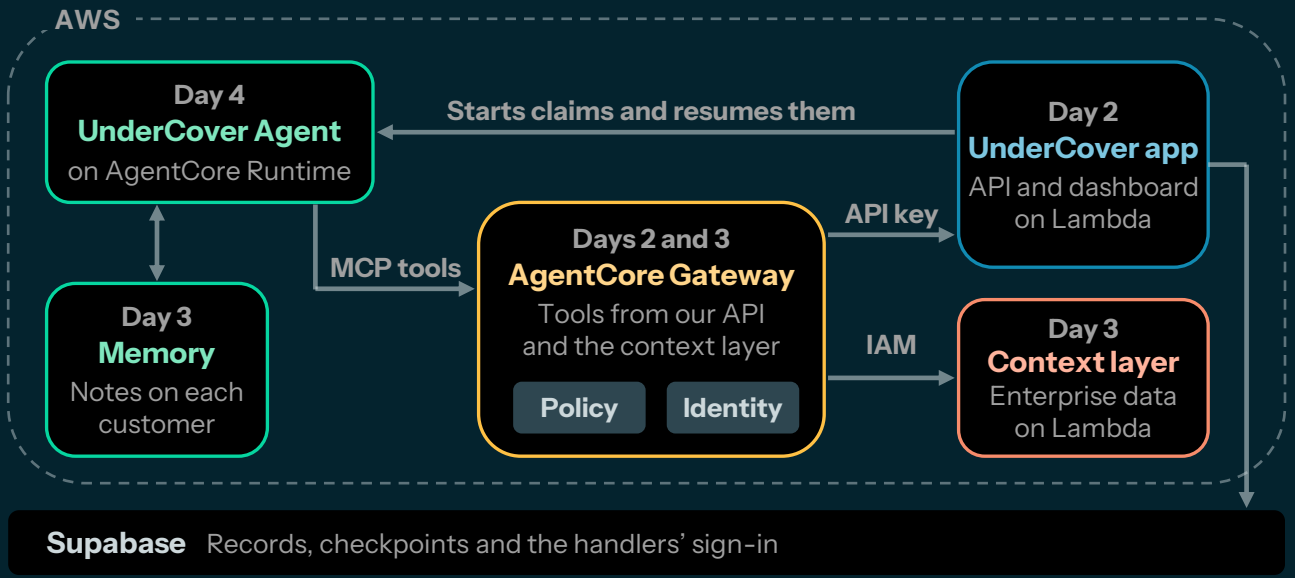
Two ways to follow the lab

Choose your depth





UnderCover by the end of Week 3



Models still come from OpenRouter

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Week 1: Become an AI Engineer

AI ENGINEER

- Path to FDE
- Frontier LLMs & Agents
- Coding Agents & ADLC
- Open-Source, Local, Token Cost
- Harness Engineering

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